

Fitness Fundamentals

A beginner's guide to the components of physical fitness and how much activity you need

What Is Physical Fitness?

Physical fitness is the ability of the body to carry out daily tasks with energy and without excessive tiredness. It reflects how well the heart, lungs, muscles, and joints work together. Fitness is built gradually through regular physical activity and is different from simply being active once in a while.

The Five Components of Physical Fitness

Health organizations generally describe fitness using five components:

1. Cardiorespiratory Endurance

The ability of the heart and lungs to supply oxygen to muscles during sustained activity, such as walking, jogging, or cycling.

2. Muscular Strength

The amount of force a muscle or muscle group can produce in a single effort, such as lifting a heavy object.

3. Muscular Endurance

The ability of a muscle to keep working over a period of time without becoming exhausted, such as carrying groceries or climbing several flights of stairs.

4. Flexibility

The range of motion available at a joint. Good flexibility supports posture and everyday movement.

5. Body Composition

The relative proportion of fat, muscle, bone, and other tissues that make up total body weight.

Why Physical Activity Matters

According to the World Health Organization (WHO) and the U.S. Centers for Disease Control and Prevention (CDC), regular physical activity is associated with:

- Improved heart and lung function
- Better weight management
- Stronger bones and muscles
- Improved mood and reduced feelings of anxiety
- Lower risk of many chronic conditions, including heart disease and type 2 diabetes

- Better sleep quality and daytime energy

How Much Activity Do Adults Need?

The WHO and CDC recommend that adults aim for at least one of the following each week:

- 150–300 minutes of moderate-intensity aerobic activity, OR
- 75–150 minutes of vigorous-intensity aerobic activity, OR
- An equivalent combination of both

In addition, muscle-strengthening activities that work all major muscle groups are recommended on 2 or more days per week. Spreading activity across the week, rather than doing it all at once, is generally encouraged.

Guidance for Other Age Groups

Children and Adolescents (5–17 years)

The WHO recommends an average of 60 minutes per day of moderate-to-vigorous activity, mostly aerobic, across the week.

Older Adults

The same general activity targets as adults apply, with an added emphasis on balance and functional movement activities to help reduce the risk of falls.

Understanding Exercise Intensity

Intensity describes how hard the body is working. A simple way to judge intensity without special equipment is the 'talk test':

- Moderate intensity: you can talk but not sing during the activity
- Vigorous intensity: you can only say a few words without pausing for breath

Warming Up and Cooling Down

A warm-up of 5–10 minutes of light activity before exercise helps prepare muscles and joints and gradually raises heart rate. A cool-down of light activity and gentle stretching after exercise helps the body return to a resting state.

Getting Started Safely

- Start at a comfortable level and increase duration or intensity gradually
- Choose activities that are enjoyable and easy to maintain long-term
- Use proper footwear and equipment suited to the activity
- Stay hydrated before, during, and after activity

- Speak with a healthcare provider before starting a new exercise program if you have an existing health condition or have been inactive for a long time

Sources

- World Health Organization (WHO) — Physical Activity Fact Sheet, and Guidelines on Physical Activity and Sedentary Behaviour
- U.S. Centers for Disease Control and Prevention (CDC) — Physical Activity Basics
- National Institutes of Health (NIH) — MedlinePlus, Fitness and Exercise
- UK National Health Service (NHS) — Fitness Guides

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